

BARSHA BHUGAI

Civil Engineer | Structural Systems & Resilient Infrastructure | Computational Engineering Enthusiast

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PROFESSIONAL SUMMARY

Civil and Rural Engineering graduate with experience in structural analysis and practical field supervision of road-upgrading and RCC bridge construction projects. Skilled in quality assurance/quality control (QA/QC), material testing, and infrastructure project monitoring for a provincial infrastructure office. Skilled in structural modeling and analysis using ETABS, SAP2000, and AutoCAD, with growing proficiency in Python-based engineering computation. Actively developing machine learning applications for structural response prediction, with a focus on seismic performance and surrogate modeling. Combines hands-on infrastructure delivery experience with a strong academic foundation in structural, transportation, and computational engineering, and is seeking a Research Assistantship or funded Master's/PhD position in Structural, Computational, or Earthquake Engineering and Smart Infrastructure.

KEY EXPERTISE

- Structural Analysis
- Finite Element Modeling
- Construction Supervision
- Infrastructure Development
- Python-based Engineering Computation
- Bridge Engineering
- Pavement Engineering
- QA/QC Management
- Machine Learning Applications

EDUCATION

Bachelor of Engineering in Civil and Rural Engineering

Pokhara, Nepal

Pokhara University | Aug 2019 – Aug 2024 | GPA: 3.26/4.0

- Relevant Coursework: Structural Analysis, RCC Design, Steel Design, Finite Element Method, RCC-Bridge Engineering, Water and Sanitary Engineering, Transportation Engineering, Hydrology and Hydraulics, Construction Management.

PROFESSIONAL EXPERIENCE

Engineer – Field Supervision & Quality Control

Feb 2026 – Jul 2026

Infrastructure Development Office (IDO), Kaski – Pokhara, Nepal

- Supervised construction activities for road-upgrading and RCC bridge projects, ensuring compliance with approved structural drawings and technical specifications.
- Evaluated detailed engineering estimates, quantity take-offs, and cost analyses (BOQ) to support budgeting for infrastructure projects valued below NPR 10 million.
- Implemented quality assurance and quality control (QA/QC) protocols through systematic field inspections and structural material testing.
- Verified construction measurements and coordinated preparation, verification, and timely processing of Interim Payment Certificates (IPCs) and final bills.
- Assessed day-to-day project progress across three concurrent contracts, maintaining technical records and construction documentation.

Civil Engineer – Supervisor

Feb 2025 – Jul 2025

Infrastructure Development Office, Kusma, Parbat, Nepal

- Supervised technical and quality-control procedures for road and bridge infrastructure projects.
- Coordinated with multidisciplinary engineering teams to support project monitoring and compliance verification.
- Prepared technical documentation and data analysis reports supporting project oversight.

Independent Technical Development & Research Preparation

Aug 2025 – Jan 2026

- Developed computational structural analysis skills using ETABS and SAP2000.
- Practiced Python-based engineering computation for structural and infrastructure problems.
- Began exploring machine learning applications for civil engineering.
- Worked on academic structural and pavement research projects.

MAJOR INFRASTRUCTURE PROJECTS – INFRASTRUCTURE DEVELOPMENT OFFICE

1. Besi Pipal Dandali – Aatighar – Mahara Mauja – Lamdanda – Sunikhola – Gehendanda Road

Road Upgrading | Sabik Pokhara-28, Kaski | 5.50 km (Ch. 0+000–5+500) | Contract ID: IDOKASKI/W/R/NCB/MY/02/079-80

- Supervised construction, QA/QC inspection, measurement verification, and compliance monitoring for this premix asphalt road-upgrading project.

2. Ghatichhina – Panchhase Road

Road Upgrading | Pokhara-28, Kaski | 2.68 km (Ch. 1+120–3+800) | Contract ID: IDOKASKI/W/R/NCB/MY/119/2080/81

- Conducted field inspection, quality control, and documentation for premix asphalt and RCC rigid pavement construction.

3. Chandikhola RCC T-Beam Bridge (Motorable Bridge)

Bridge Construction | Pokhara-28, Kaski | RCC T-Beam, span 15 m | Contract ID: IDOKASKI/W/BRG/NCB/MY/126/2080/081

- Supervised construction activities, verified structural measurements, and ensured compliance with design specifications and quality standards.

RESEARCH & COMPUTATIONAL DEVELOPMENT

Machine Learning-Based Surrogate Modeling for Rapid Prediction of Nonlinear Seismic Response of Reinforced Concrete Structures

Ongoing

- Developing machine learning frameworks for predicting nonlinear structural responses of reinforced concrete structures.
- Using simulation-based engineering datasets and structural design parameters to inform model development.
- Exploring surrogate modeling approaches to reduce the computational cost of nonlinear seismic analysis.
- Investigating applications in seismic analysis and resilient structural design.

SELECTED ACADEMIC & COMPUTATIONAL PROJECTS

Computational Modeling of Bridge Structures – RCC T-Beam Bridge Analysis

May 2025 – Jul 2025

- Performed finite element modeling in ETABS to evaluate structural response of a 25.5 m RCC T-Beam bridge, including load distribution, shear force, and bending moment analysis.

Pavement Systems Performance Study – Road Upgrade Project

Mar 2025 – May 2025

- Compared flexible (AC) and rigid (RCC) pavement systems for a 6-km mountain road, developing an optimization framework based on soil, traffic, and environmental considerations.

Infrastructure Planning & Design – Rural Road Development DPR

Mar 2024 – Jul 2024

- Prepared a Detailed Project Report (DPR) for a 3-km mountain road upgrade using AutoCAD and Smart Road for infrastructure planning and design visualization.

Sustainable Pavement Materials – Plastic Waste in Bituminous Road Construction

Jul 2023 – Sep 2023

- Investigated mechanical properties of plastic-modified versus conventional bitumen through experimental testing and comparative analysis.

TECHNICAL SKILLS

- Structural & Engineering Software: ETABS, SAP2000, AutoCAD, Civil 3D, Smart Road, SketchUp
- Programming & Data Science: Python, Machine Learning, Jupyter Notebook, LaTeX, Advanced Excel
- Engineering Testing & Field Skills: CBR, Proctor Test, Atterberg Limits, Marshall Stability, Bitumen Testing, Concrete Strength Testing, NDT

RESEARCH INTERESTS

- Machine learning-assisted structural analysis
- Surrogate modeling for nonlinear FEM simulations
- Structural health monitoring
- Earthquake-resistant infrastructure
- Smart and resilient infrastructure systems
- Data-driven civil engineering

CERTIFICATIONS & ACHIEVEMENTS

- Certified National Chess Arbiter (FIDE)
- Continuous learning in Python and machine learning applications.

REFERENCES

Prof. Dr. Tek Raj Gyawali, Professor, School of Engineering, Pokhara University, Nepal – tekg@pu.edu.np

Krishna Acharya, Senior Divisional Engineer, Infrastructure Development Office (IDO), Pokhara, Kaski, Nepal